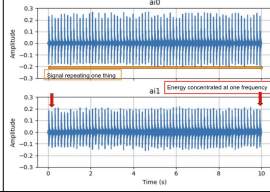
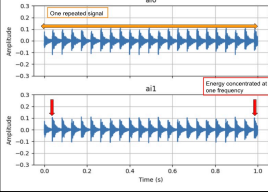
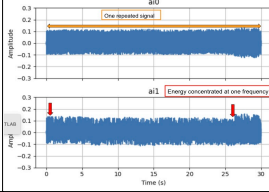
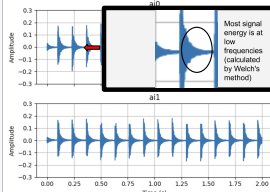
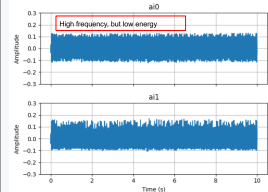
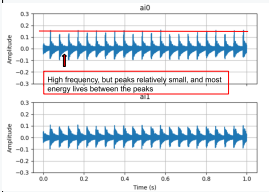
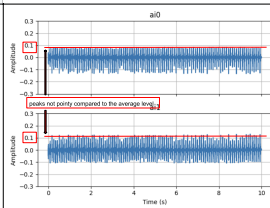
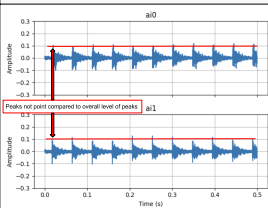
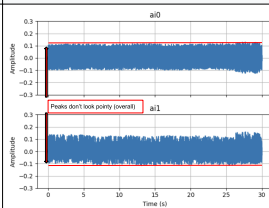
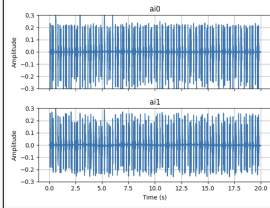
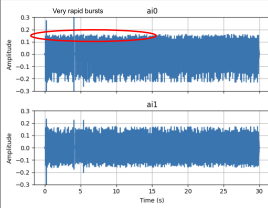
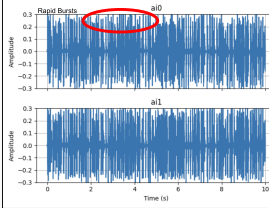
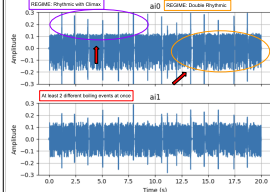
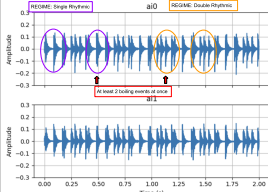
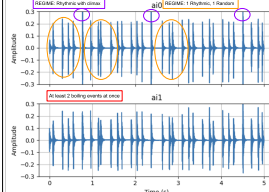
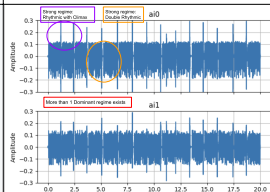
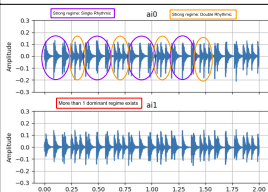
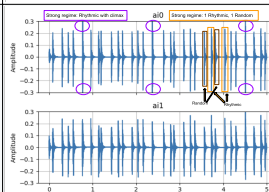


Subcluster #	Feature Name	Previous or Created	Time Detrending	Intuitive Meaning	How it's calculated	Mean of Feature in Cluster	Signal Example 1	Signal Example 2	Signal Example 3
1	Spectral Entropy	Previous		How spread out the signal's energy is across frequencies (high = more spread out)	Calc power spectral density with Welch's method, normalize it, find Shannon entropy	Low Spectral Entropy			
1	Spectral Centroid	Previous		Where the average frequency of the signal lies (low frequencies = low centroid)	Weighted average of frequencies	Low Spectral Centroid			
1	Crest factor	Created		How extreme the peaks are relative to the overall "energy" of the signal. High crest factor = short, sharp spikes	Peak amplitude divided by root mean square of the signal	Low Crest Factor			
2	Burstiness	Created		Measures how irregular the signal is. High burstiness = periods of rapid activity followed by slow periods	std deviation of inter peak arrivals divided by mean of inter peak arrivals	High Burstiness			
2	num_boilings	Previous	Yes	Number of distinct patterns of boiling found	From previous team's code	High Number of Boiling patterns			
2	regime_dominance	Created		Proportion of time spent in each regime, which regime dominates	Find regimes using previous team's code, then calc/find the proportion of the dominant regime.	Low Regime Dominance			

Subcluster #	Feature Name	Previous or Created	Time Detrending	Intuitive Meaning	How it's calculated	Mean of Feature in Cluster	Signal Example 1	Signal Example 2	Signal Example 3
3	Std_dev_time_diff	Previous		Std deviation of the inter peak times	Find the peaks, find time differences between the peaks. Find std. deviation of those differences.	Very High Std Dev of Time Between Peaks			
3	kurtosis	Created		Measures how heavy the tails of the distribution are. High kurtosis = occasional very high peaks	The fourth standardized central moment	High Kurtosis			
3	crest factor	Created		How extreme the peaks are relative to the overall "energy" of the signal. High crest factor = short, sharp spikes	Peak amplitude divided by root mean square of the signal	High Crest Factor			
4	skewness	Created		Assymetry of the signal's distribution around its mean (positive skew = occasional high spikes)	The third standardized central moment	High skewness			
4	Mean_time_diff	Previous	Yes	Average time interval between peaks (mean of inter peak times)	Find the peaks, find time differences between the peaks. Find mean of those differences.	High mean time diff.			
4	Median_time_diff	Previous	Yes	Median of the time differences (median of inter peak times)	Find the peaks, find time differences between the peaks. Find median of those differences.	High median time diff.			

Subcluster #	Feature Name	Previous or Created	Time Detrending	Intuitive Meaning	How it's calculated	Mean of Feature in Cluster	Signal Example 1	Signal Example 2	Signal Example 3
5	Peaks_per_second	Previous	Yes	How many peaks appear per second on average	Total number of peaks divided by length of time	Low peaks per second			
5	regime_dominance	Created		Proportion of time spent in each regime, which regime dominates	Find regimes using previous team's code, then calc/find the proportion of the dominant regime.	High regime dominance			
5	num_boilings	Previous	Yes	Number of distinct patterns of boiling found	From previous team's code	Low Number of Boiling patterns			

Note: we're exploring different features for subclustering (and subcluster groupings), and may present better suggestions on thursday!

REGIME LABELS FROM PREVIOUS GROUP:		1 = Rhythmic Regime	2 = Rhythmic Regime	3 = Not a Rhythmic Regime	4 = Rhythmic Regime	5 = Not a Rhythmic Regime	6 = Rhythmic Regime	7 = Not a Rhythmic Regime
Regime 1: Single Rhythmic	Regime 2: Double Rhythmic	Regime 3: Random	Regime 4: Rhythmic With Climax	Regime 5: Noise	Regime 6: 1 Rhythmic, 1 Random	Regime 7: Rapid		
Signal for MATLAB 1:02 PM Thu, Nov 7, 2024 Run8 .csv 	Signal for MATLAB 1:00 PM Fri, Jun 29, 2024 Run8 .csv 	Signal for MATLAB 1:10 PM Thu, Aug 22, 2024 Run8 .csv 	Signal for MATLAB 2:15 PM Thu, Oct 31, 2024 Run8 .csv 	Signal for MATLAB 1:19 PM Tue, Oct 29, 2024 Run8 .csv 	Signal for MATLAB 3:00 PM Tue, Oct 22, 2024 Run3 .csv 	Signal for MATLAB 2:00 PM Tue, Oct 22, 2024 Run3 .csv 		
Signal for MATLAB 4:57 PM Tue, Oct 1, 2024 Run11 .csv 	Signal for MATLAB 3:19 PM Wed, Sep 18, 2024 Run10 .csv 	Signal for MATLAB 1:20 PM Thu, Aug 22, 2024 Run8 .csv 		Signal for MATLAB 1:07 PM Fri, Jun 14, 2024 Run1 .csv 	Signal for MATLAB 1:00 PM Tue, Oct 29, 2024 Run1 .csv 			